

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-1038
Project Name	Auto Ticket Classification Using Flow Designer
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	ServiceNow Service Portal / Incident Workflow Form	Provides the user-facing form for students and teachers to enter caller details and a short ticket description. Keeps ticket submission simple and structured.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	ServiceNow Flow Designer	Acts as the automation engine. It triggers when a ticket is created, evaluates predefined keywords, updates Category and Subcategory, and sends the confirmation email.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	ServiceNow Incident Workflow Table	Stores ticket records, caller information, short descriptions, categories, subcategories, and reference numbers. The project uses the custom Incident WorkFlow table (u_incident_workflow).

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
2	Backend / Server-Side	ServiceNow Flow Designer	Visual, no-code workflow processing for automatic ticket classification, record updates, and notification.
3	Database / Data Storage	ServiceNow Tables / Data Dictionary	Stores ticket data and enforces dependent Category–Subcategory choices so invalid combinations are prevented.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	ServiceNow Cloud Platform	Provides the hosted ServiceNow environment for the application, Flow Designer automation, ticket records, and email services.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	ServiceNow Update Sets / XML Export	Packages configuration, table changes, dictionary settings, and Flow Designer changes for controlled migration between Development, Test, and Production instances.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
5	Version Control & CI/CD	ServiceNow Update Sets / XML Export	Supports portable deployment of the completed project configuration as an Update Set XML package.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	ServiceNow REST APIs / IntegrationHub	Provides integration capability for external services when required. The core classification and notification workflow is implemented natively in ServiceNow.